

PROJECT REPORT

Team ID: LTVIP2025TMID48073

1. INTRODUCTION

Project Overview: Comprehensive House Price Analysis

This project utilizes Tableau to create a **visual analytics dashboard** that helps users explore how various factors affect house prices. Drawing from a cleaned dataset (Transformed_Housing_Data2.csv), the dashboard includes multiple views to analyze relationships between housing features and their impact on sales prices.

Key visuals include:

- **Sales Price vs Flat Area**
- **Sales Price vs Waterfront View**
- **Distribution of House Age by Renovation Status**
- **Sales Price vs Age of House**
- **Total Sales by Years Since Renovation**
- Summary stats: total houses, average price, total basement area

The dashboard is designed to make **housing trends easily interpretable** for real estate professionals, homeowners, buyers, and urban planners.

Project Purpose

The purpose of this project is to:



1. Enable Data-Driven Housing Insights

- Identify how **flat area, waterfront presence, age of house, and renovations** influence the **sale price**.
- Visualize average price trends to support buyer or investor decisions.



2. Improve Decision-Making for Stakeholders

- **Buyers** can compare what features raise house prices.
- **Sellers** can estimate the value added by renovations.
- **Policy makers** can observe trends for planning or subsidies.



3. Detect Key Patterns

- Analyze if newer or recently renovated homes sell better.
- Spot areas of opportunity (e.g., if houses with certain features are undervalued).



4. Summarize Critical Metrics

- Track dataset size and aggregated metrics (total area, avg. price).
- Understand the **distribution of housing features** quickly through color-coded charts.

IDEATION PHASE

Problem Statement

Customer Problem Statement :

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	A home buyer/investor wanting to make informed decisions, interested in area, price, and renovation impacts
I'm trying to	Understand how factors like flat area, renovation status, and waterfront view affect house prices to plan purchases
but	The data is scattered, unorganized, and hard to analyze visually, making it difficult to identify trends and patterns
because	There is no clear, interactive, visual tool to explore how these factors influence house prices in one place
which makes me feel	Confused, overwhelmed, and lacking confidence in making high-value property decisions

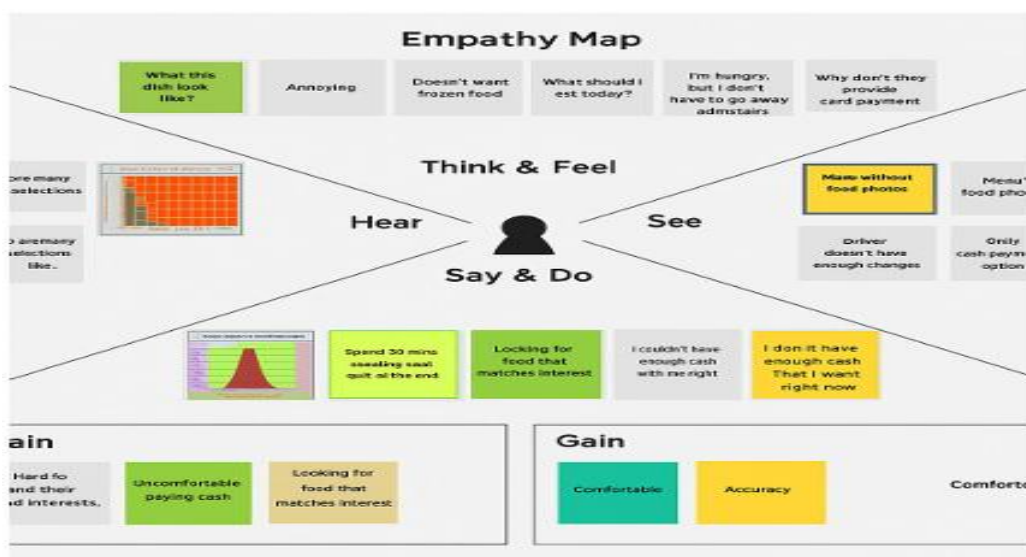
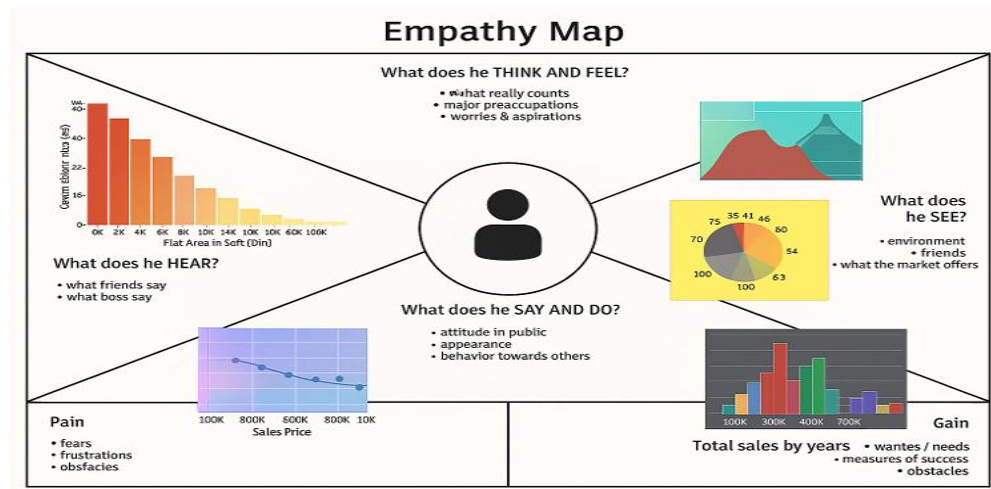


Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A home buyer/investor wanting clear insights on pricing factors	Understand how flat area, renovation, and waterfront view affect house prices	The data is scattered and hard to analyze visually	There is no interactive tool to view these factors clearly	Confused and hesitant in making property decisions
PS-2	A real estate analyst preparing market insights	Visualize pricing trends and identify patterns for client reporting	Existing tools are complex and lack dynamic filtering	Available data lacks clear, interactive graphical representation	Frustrated and delayed in preparing reports efficiently

Empathy Map Canvas

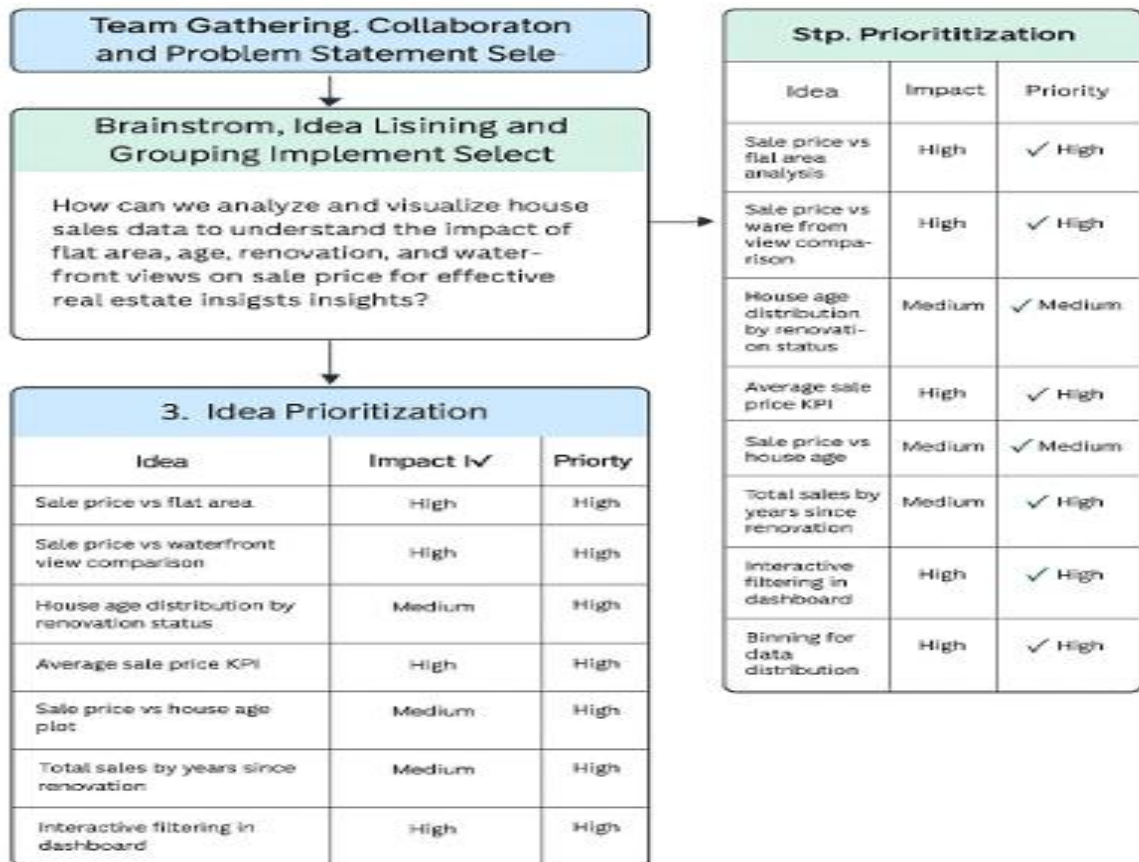
An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It is a useful tool to help teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Brainstorming

Brainstorm & Idea Prioritization Based on Comprehensive House Price lalysis



Step-2: Brainstorm, Idea Listing and Grouping

1

Brainstorm
Write down any ideas that come to mind that address your problem statement.
⌚ 30 minutes

Tip

You wrote a good sticky note to present ideas.

Amar

Average Size

Yukesh

Average size

Person 3

Person 5

Person 6

Person 8

2

Group Ideas
Take turns sharing your ideas while clustering similar, or related notes as you go. In the two 10 minutes, give each cluster a sentence label. If a cluster is bigger than six sticky notes, try and see if y
⌚ 20 minutes

Person 4

Make a conditional table to why or exactly when area to add meaning from. Important make features, or more better note.

Step-3: Idea Prioritization

Importance

Feasibility

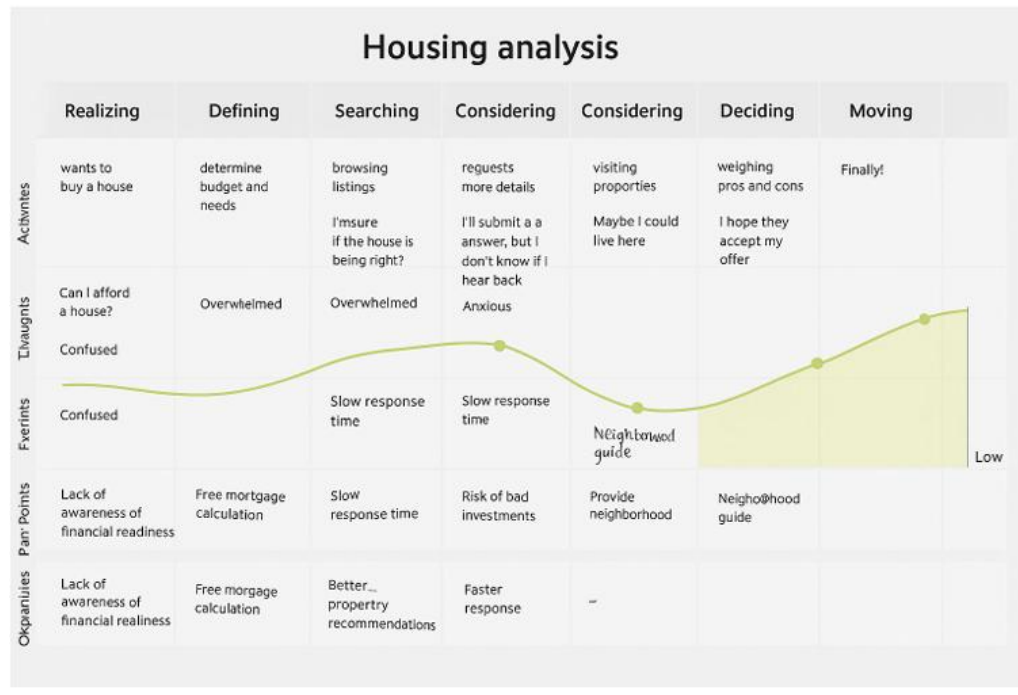
Heart icon at origin

Idea Cards:

- Total Sales by Years Since Renovation (Purple)
- Sale Price vs House Age Plot (Purple)
- Sale Price vs Flat Area Analysis (Green)
- Sale Price vs Waterfront View Analysis (Green)
- Avg. Sale Price KPI Block (Green)
- Interactive Filtering in Dashboard (Green)
- House Age Dist. by Renovation Status (Green)

REQUIREMENT ANALYSIS

Customer Journey map:



Solution Requirement:

Functional Requirements: Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Upload and Integration	Upload dataset from user system Connect Tableau to data source
FR-2	Data Exploration and Filtering	Apply filters on flat area, renovation status, house age, sale price
FR-3	Visualization of Housing Metrics	Show charts: Histogram, Pie Chart, Scatter Plot, Stacked Bar
FR-4	Summary Metrics	Show KPIs: Average Sale Price, Total Count, Basement Area
FR-5	Recommendation Generation	Suggest ideal house types (size, renovation status, location)
FR-6	User Storyline & Dashboard View	Create Tableau Story with sequential insights

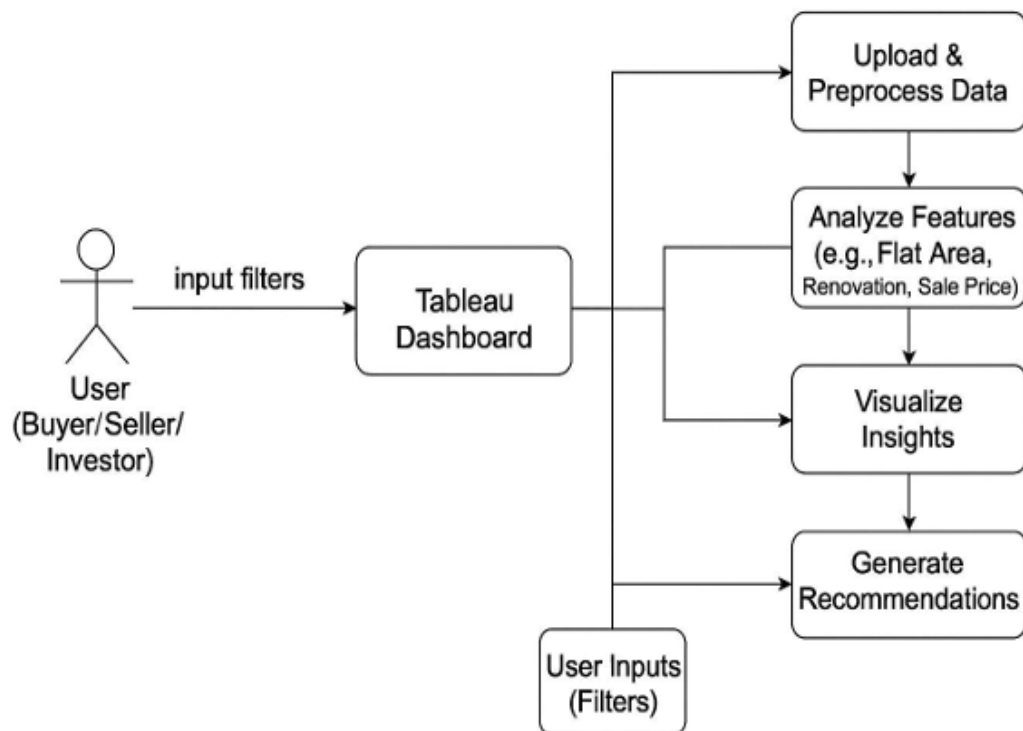
Non-functional Requirements: Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple dashboard navigation and chart interactivity for users
NFR-2	Security	Role-based access and dataset protection
NFR-3	Reliability	Dashboard must refresh with new uploads without errors
NFR-4	Performance	Visuals load in under 3 seconds for datasets < 25,000 rows
NFR-5	Availability	Accessible 24/7 via Tableau Public or integrated portal
NFR-6	Scalability	Can support additional visualizations and filters without redesign

Data Flow Diagram:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DATA FLOW DIAGRAM (LEVEL 0 – CONTEXT LEVEL)



User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Analyst/User	Data Upload & Exploration	USN-1	As a user, I want to upload housing data to explore trends using Tableau dashboards.	Data upload succeeds, visualization is populated	High	Sprint-1
Analyst/User	Filtering and Custom View	USN-2	As a user, I want to filter the charts by area, renovation, and age of house.	Charts change based on selected filters	High	Sprint-1
Buyer/Investor	Understand Sale Price Trends	USN-3	As a user, I want to see what house features most impact price (area, waterfront, renovation).	Trend line and chart visuals show this clearly	High	Sprint-1
Developer	Insights Summary	USN-4	As a user, I want a KPI dashboard that summarizes record count, avg. price, and basement area.	Display of 3 clean KPIs on top of the dashboard	Medium	Sprint-2
Buyer	Get House Recommendations	USN-5	As a buyer, I want recommendations for house types that suit my criteria.	Data-driven suggestions visible based on selected input	Medium	Sprint-2

Technology Stack:

Technical Architecture:

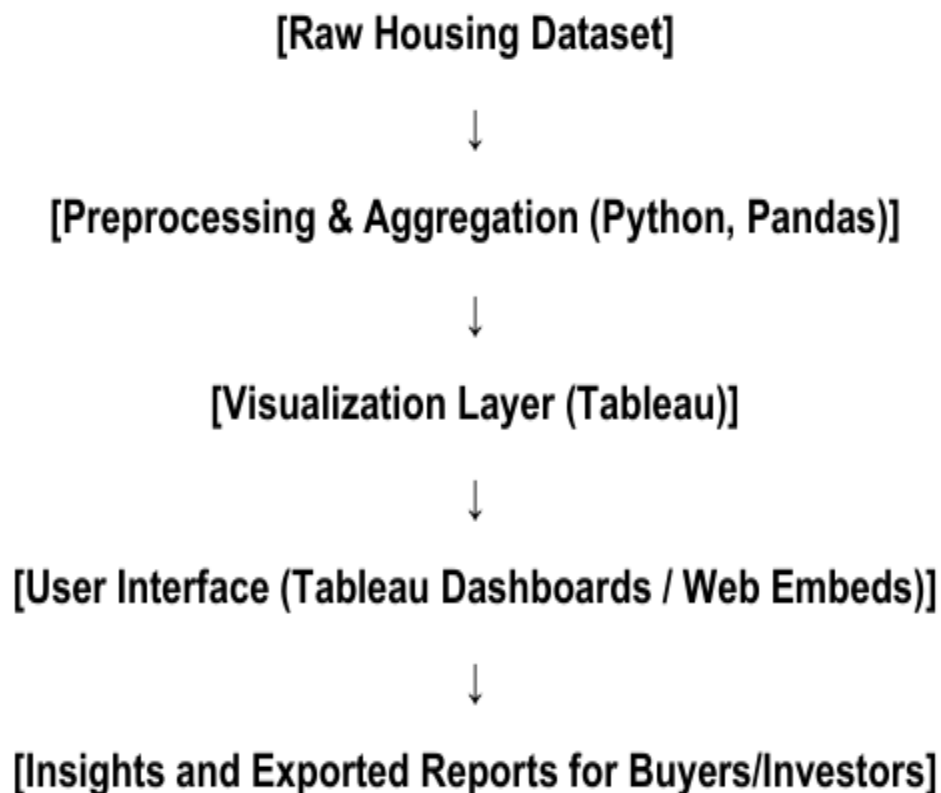


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Visualization dashboards	Tableau Public, HTML Embed
2	Application Logic-1	Data processing and metric generation	Python (Pandas, NumPy)
3	Application Logic-2	KPI calculations, trend modeling	Tableau Calculated Fields
4	Application Logic-3	Recommendation logic based on house features	Custom filters in Tableau
5	Database	Source housing dataset	CSV file / SQLite
6	Cloud Database	Optional for larger housing data	Google Sheets / AWS S3
7	File Storage	Local and online Tableau extracts	Local File System, Tableau Online
8	External API-1	Location metadata (optional)	Google Maps API (optional)
9	External API-2	Weather impact analysis (optional extension)	OpenWeather API
10	Machine Learning Model	Forecasting renovation-based value uplift (future scope)	Scikit-learn (optional)
11	Infrastructure	Tableau Public / Local PC for deployment	Tableau Desktop, Web Browser

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

PROJECT DESIGN:

Problem Solution Fit:

Problem Statement Many home buyers, real estate investors, and property developers struggle to understand how house characteristics like flat area, age, renovation history, and waterfront views influence sale price. Without data-backed insights, they risk overpaying, underpricing, or missing investment opportunities.

Purpose:

A Tableau-based visual analytics dashboard that allows users to explore:

- How different housing features impact sale prices
- Market trends and buyer preferences through visualizations (histograms, pie charts, scatter plots, etc.)
- Summary KPIs (average sale price, basement area, property count)
- Data-driven recommendations for renovation timing, pricing strategy, and target house types

Template:

PROBLEM STATEMENT
Many home buyers, real estate investors, and property developers struggle to understand how various car characteristics like—flat area, age, renovation history, and waterfront views—influence sale price. Without data-backed insights, they risk overpayin, underpricing, or missing investment opportunity.

PROPOSED SOLUTION
A Tableau-based visual analytics dashboard that allows users to explore: • How different housing features impact sale prices • Market trends and buyer preferences through visalizations (histograms, pie charts, scatter -plots, etc.) • Summary KPIs: Average sale price, basement area; proright riggers, and target husse

PROBLEM-SOLUTION FIT	
Category	Fit Insights
Customer Behavior	Buyers rely on trends and guesswork entstedof insights
Core Problem	Lack a clear understanding of what affects house
Existing Alternatives	Manual filtering on listing sites, Excel reports
	interactive visuals, real-time filters, clearer patterns
Solution Advantage	Informed decisions, optimized pricing, smarter investments
VALUE DELIVERED	• Reduces uncertainty in real estate buying/sell • Empowers users with self-service insights • Saves time avodd costly guesswork

Proposed Solution:

Proposed Solution Template: Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Home buyers, sellers, and investors lack clear insights into how house features like flat area, age, renovation, and view affect sale prices.
2.	Idea / Solution description	A Tableau-based dashboard that visually analyzes feature-driven pricing behavior.
3.	Novelty / Uniqueness	Unlike static reports, our solution offers interactive filtering, data stories , and visual recommendations based on user-selected inputs .
4.	Social Impact / Customer Satisfaction	Buyers make smarter investments, sellers price homes more accurately, and developers understand renovation impact—resulting in increased satisfaction and trust.
5.	Business Model (Revenue Model)	Freemium model: core dashboard is free, with op consulting on pricing strategy as premium service
6.	Scalability of the Solution	The solution can easily scale across different cities, property types, and datasets. More features like location intelligence or ML forecasting can be integrated.

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions.

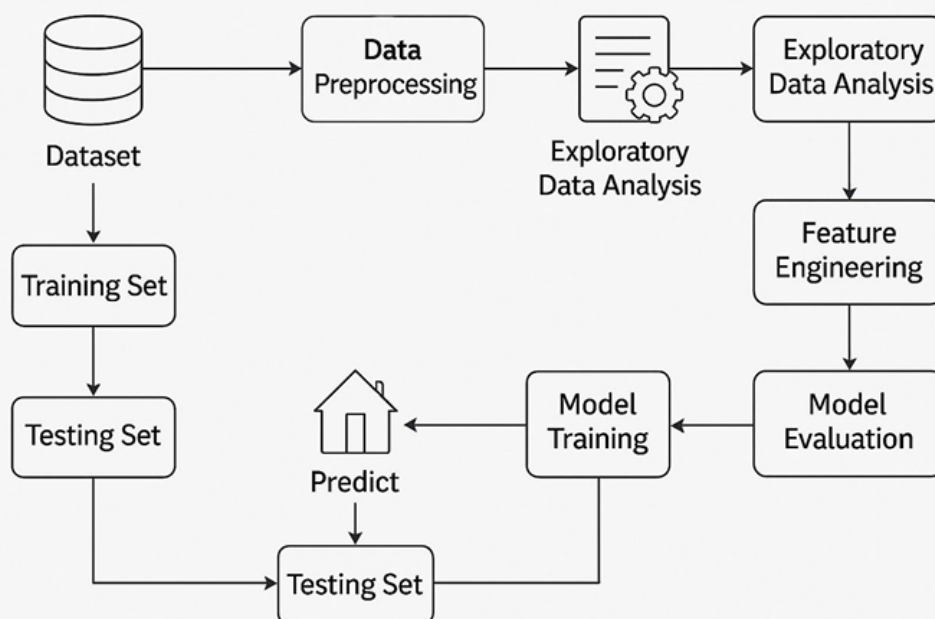
Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Proposed Solution – House Price Analysis

1	Problem Statement (Problem to be solved)	Home buyers, sellers, and investors lack clear insights into how house features like flat area, age, renovation, and view affect sale prices.
2	Idea / Solution description	A Tableau-based dashboard that visually analyzes housing data to highlight sale trends, key KPIs, and feature-driven pricing behavior
3	Novelty / Uniqueness	Unlike static reports, our solution offers <i>interactive filtering</i> , data stories, and visual recommendations based on user-selected inputs
4	Social Impact / Customer Satisfaction	Buyers make smarter investments, sellers price homes more accurately, and developers understand renovation impact—resulting in increased satisfaction and trust
5	Business Model (Revenue Model)	Freemium model: core dashboard is free, with options for personalized reports, predictive insights, or consulting on pricing strategy as premium services

House Price Analysis



PROJECT PLANNING & SCHEDULING:

Project Planning :

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Preparation	USN-1	Collect housing dataset	2	High	member3
Sprint-1	Data Preparation	USN-2	Load dataset into Python or Tableau	1	High	member3
Sprint-1	Data Cleaning	USN-3	Handle missing values	3	Medium	member3
Sprint-1	Data Transformation	USN-4	Encode categorical variables	2	Medium	member3
Sprint-2	Visualization (Exploratory)	USN-5	Create KPI cards (average price, total count, etc.)	2	High	member3
Sprint-2	Visualization	USN-6	Build histogram (Sale Price vs Area)	2	High	member3
Sprint-2	Visualization	USN-7	Pie chart (House Age & Renovation)	2	Medium	member3
Sprint-2	Visualization	USN-8	Scatter plot with trend line (Age vs Sale Price)	3	Medium	member3
Sprint-2	Visualization	USN-9	Stacked bar chart (Renovation Age vs Sale Price)	3	Medium	member3
Sprint-3	Storytelling & Reports	USN-10	Create Tableau story views	2	High	member3
Sprint-3	Reports	USN-11	Document insights and export as PDF	2	Medium	member3
Sprint-3	Design & Maps	USN-12	Generate empathy & customer journey map	3	Medium	member3
Sprint-3	Architecture & Logic	USN-13	Create DFD and tech stack visualizations	3	High	member3

print	Total Story Points	Duration	Sprint Start Date	Sprint End Date	Story Points Completed	Sprint Release Date
Sprint-1	8	5 Days	15 June 2025	20 June 2025	8	20 June 2025
Sprint-2	12	5 Days	20 June 2025	25 July 2025	12	25 June 2025

Project Tracker, Velo
Chart: (4 Marks)

Sprint-3	12	5 Days	25 June 2025	30 June 2025	12	30 June 2025
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FUNCTIONAL AND PERFORMANCE TESTING:

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Dataset: Transformed_Housing_Data2.csv Total Records: 21,609 Key Fields: Sale Price, Flat Area, Waterfront View, Age of House, Years Since Renovation
2	Data Preprocessing	- Null values checked and handled - Binning on Flat Area and Sale Price - Derived Years Since Renovation - Outlier removal for extreme sqft
3	Utilization of Filters	- Sale Price (bin) - Years Since Renovation - Count of Sale Price (slider) - Interactive filtering across graphs
4	Calculation Fields Used	- Years Since Renovation = [Year Sold] - [Year Renovated] - Average Sale Price - Sale Price and Flat Area binning - Sum and Count of Sale Price

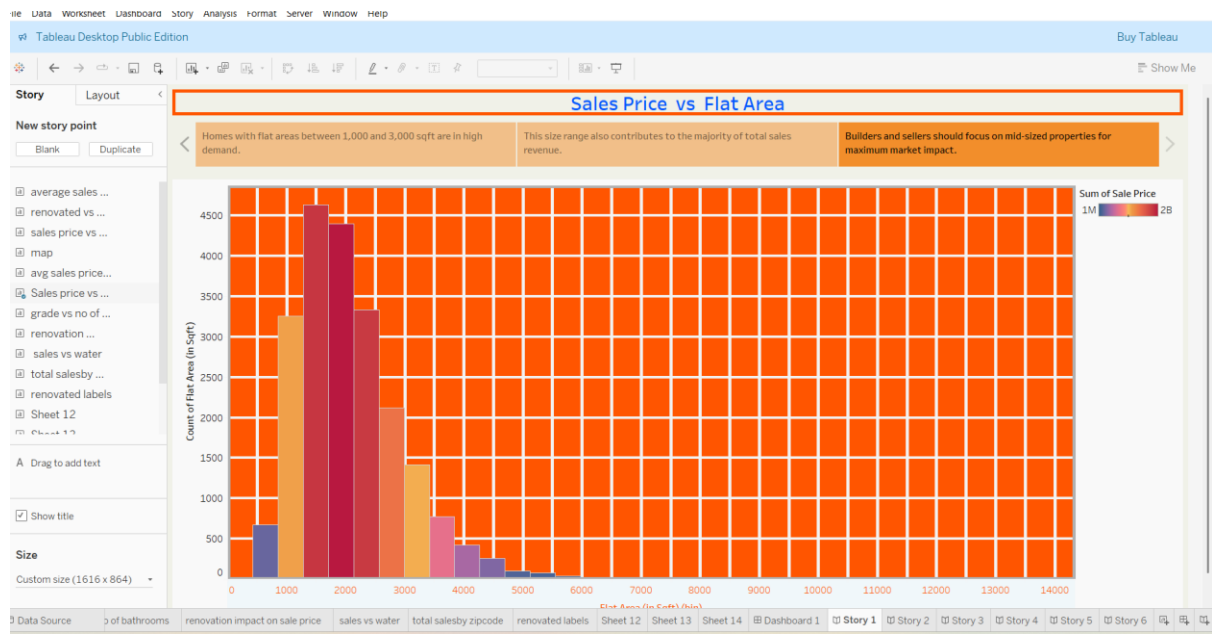
5	Dashboard Design	No. of Visualizations / Graphs: 6 1. Sales Price vs Flat Area (Histogram) 2. Sales Price vs Waterfront View (Density Curve) 3. Distribution of House Age by Renovation Status (Pie) 4. Sales Price vs Age of House (Scatter) 5. Total Sales by Years Since Renovation (Histogram) 6. KPI Block (Count, Average Price, Total Area)
6	Story Design	No. of Visualizations / Graphs: 6 (integrated within the dashboard for narrative flow)

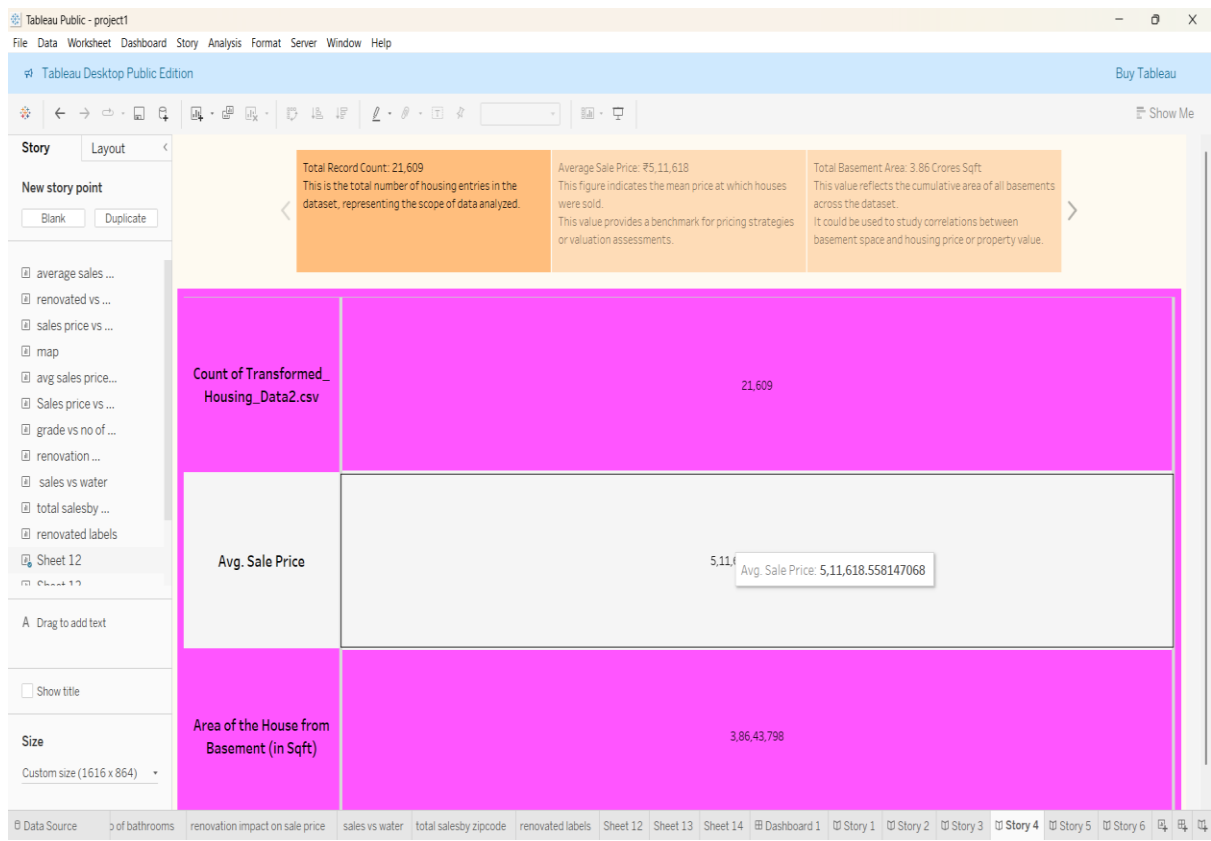
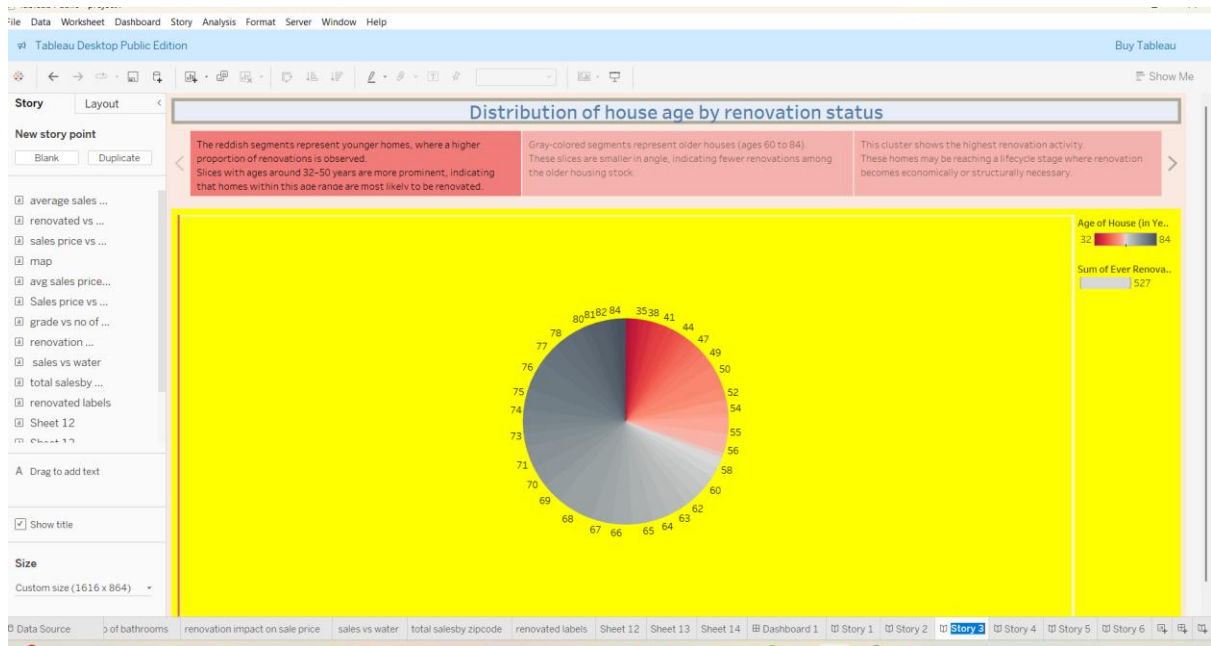
Results:

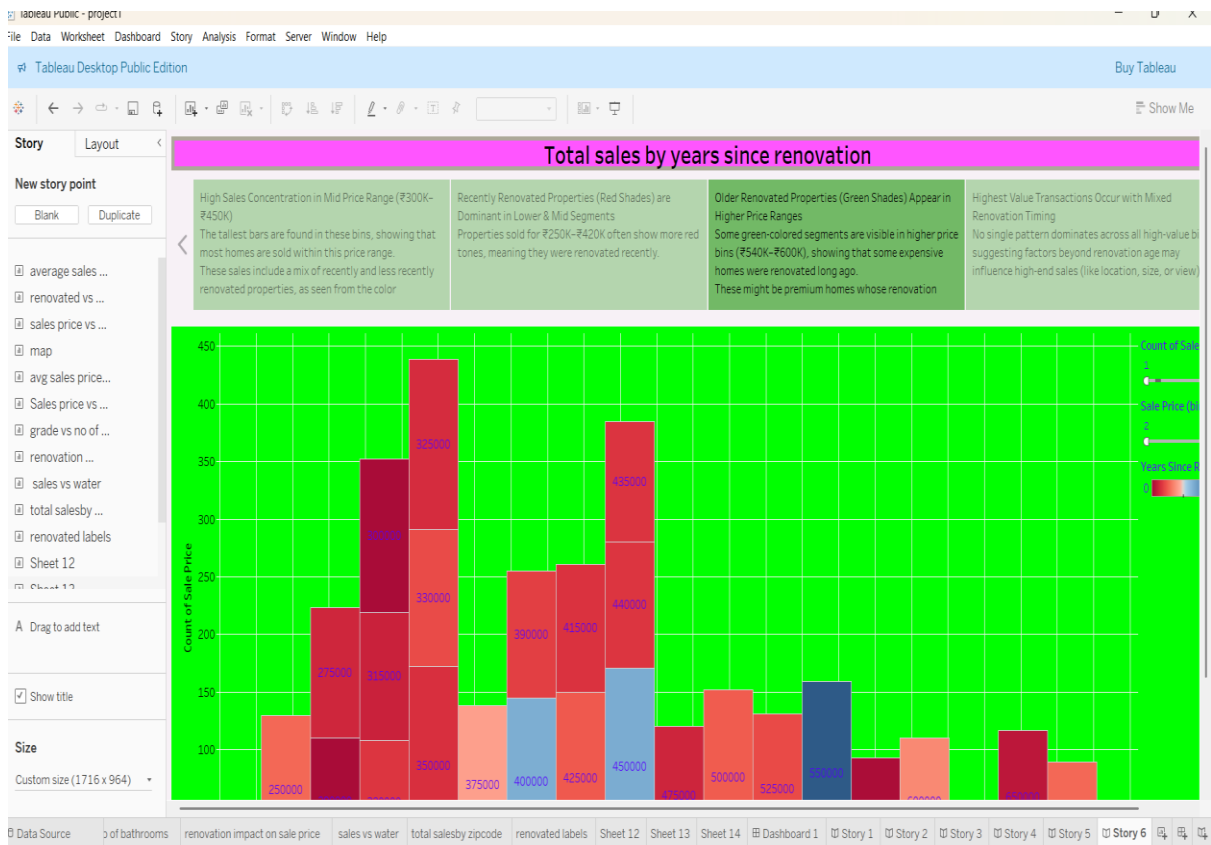
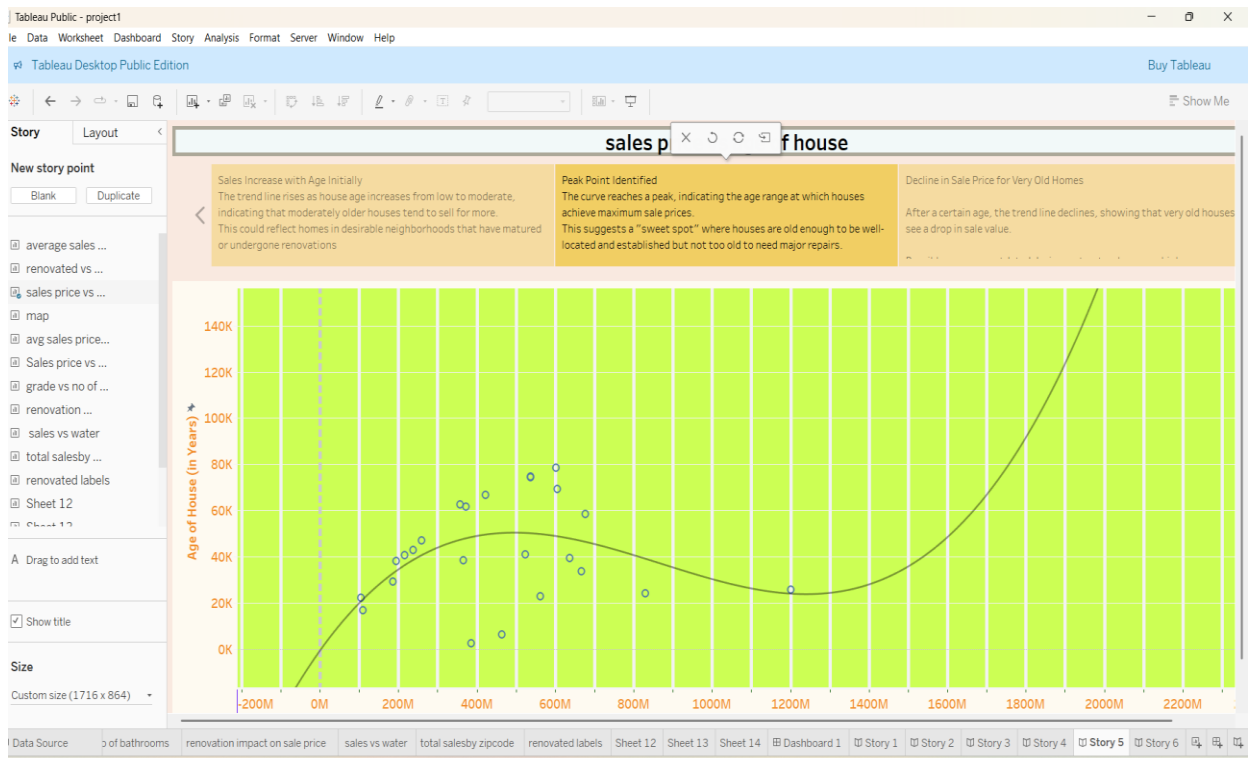
Dashboard:



Story:







Advantages:

1. Data Visualization for Decision-Making

- Easy-to-understand dashboards (bar charts, pie charts, scatter plots, etc.) enable quick insights into house price factors like:
 - Flat area
 - Waterfront view
 - Renovation history
 - House age

2. Customer-Centric Design

- Empathy maps and customer journey maps capture buyer frustrations, needs, and behaviors (e.g., cash payment pain points, visualizing food/housing features), helping improve UX and service design.

3. Structured Development with User Stories

- Use of user stories ensures functional requirements are met (registration, login, data access), helping align development with real user goals.

4. Predictive Trends

- Visuals such as **sales price vs. house age** help identify how renovations or house age influence value, guiding investment decisions.

5. Comprehensive Data Coverage

- The dataset includes 21,609 entries, providing enough data for statistically significant insights.

6. Easy Sharing (e.g., Tableau Public, GitHub)

- Tableau allows interactive dashboards to be published or shared as .twbx, PDF, or image format, ensuring accessibility and collaboration.

Disadvantages / Limitations:

1. Static Visuals for Some Users

- If not using Tableau Public, viewers see only static .png or .pdf versions and miss interactivity (e.g., filtering, tooltip info).

2. No Real-Time Data Integration

- The dashboard is based on static, preprocessed data (CSV) rather than real-time housing feeds or APIs, which limits up-to-date analysis.

3. Complexity for Non-Technical Users

- Understanding scatter plots and regression curves may be challenging for some non-analyst users (e.g., first-time home buyers).

4. Missing Geospatial Visualization

- Although there's a “map” sheet in Tableau, the screenshot doesn't show geographic insights like zip-code level distribution, which could add more value.

5. Design Consistency & UX

- The dashboard uses very bright and saturated colors (pink, green, yellow) that can be visually overwhelming and less professional for stakeholders.

6. Limited Automation

- There's no integration yet with scheduling or alerting mechanisms (e.g., for monthly trend updates or anomaly detection).

Conclusion :

The **Comprehensive House Price Analysis project** effectively leverages Tableau's powerful visualization capabilities to uncover insights in the housing market. Through detailed dashboards, we analyze how variables like **flat area**, **waterfront views**, **renovation history**, and **house age** impact **sales prices**.

Supporting tools like **customer journey maps** and **empathy maps** offer a user-centered lens into the homebuyer's experience—highlighting common frustrations (e.g., payment issues, lack of photo references) and emotional triggers during the decision-making process. This dual approach—data analytics plus user empathy—positions the project as both **technically insightful** and **customer-aware**.

Moreover, the integration of **user stories** and **data flow diagrams** ensures structured development aligned with user needs, enhancing usability and maintainability.

Future Scope

1. Geospatial Analysis

- Incorporate **maps by ZIP code** or city for location-based insights (e.g., where renovations raise price most).
- Add choropleth mapping for better visual clarity.

2. Real-Time Data Integration

- Connect to **live housing market APIs** (e.g., Zillow, Redfin) for dynamic dashboards.
- Enable auto-refresh in Tableau for up-to-date analysis.

3. Predictive Modeling & ML

- Apply regression, decision trees, or time-series forecasting to predict future housing trends.
- Suggest optimal renovation strategies based on ROI.

4. Enhanced User Personalization

- Use historical interaction data to personalize dashboards for different users (e.g., first-time buyers vs investors).

5. Mobile Optimization

- Redesign dashboards for **mobile-first viewing**, using Tableau's "Phone Layout" feature more extensively.

6. Web Integration & Embedding

- Embed dashboards into real estate websites or mobile apps with authentication.
- Enable secure user filtering (e.g., by user account or preferred location).

7. User Feedback Loop

- Add feedback forms directly in the dashboard.
- Use insights to improve UX based on real usage patterns.

APPENDIX:

Dataset link:

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

GitHub link:

[Upload files · Anitha123-allam/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau](#)

Video demo link:

https://drive.google.com/file/d/1gbX-LLdlk17j3ZhHT7FHkUepY-DBFNVX/view?usp=drive_link